STATEMENT OF PURPOSE

Amazon ML Summer School 2026 Submission

My technical journey in AI/ML is driven by a fascination with high-dimensional data and a desire to build compute-efficient, localized, and clinically viable intelligent systems. Combining frameworks in electronics and computer science, I have focused my undergraduate research on translating complex theoretical architectures into robust, functional codebases.

In the medical imaging domain, I have focused heavily on computer vision challenges where structural precision is critical. I implemented a 3D Convolutional Neural Network (3D CNN) utilizing the EchoNet-Dynamic dataset to predict Ejection Fraction directly from echocardiogram videos, optimizing spatial-temporal feature extraction. Expanding on segmentation, I engineered an end-to-end medical segmentation pipeline in PyTorch for delineating low-grade gliomas in brain MRIs. By integrating a Convolutional Block Attention Module (CBAM) into stripped EfficientNetV2 and MobileNetV3 backbones, coupled with a custom transposed-convolutional decoder, I built a dual-model ensemble that successfully handles extreme pixel-class imbalances.

Beyond computer vision, I am highly interested in sequence modeling and multi-agent orchestration. I developed a financial forecasting pipeline using Long Short-Term Memory (LSTM) networks with a Flask backend to model temporal dependencies. Driven by a "local-first" philosophy to ensure data privacy, I also architected an autonomous email assistant using LangGraph and Ollama (Llama 3.2), implementing deterministic tool-calling state machines that run entirely on consumer hardware.

Despite these practical implementations, my work has primarily existed within localized environments and structured datasets. A significant gap remains in my understanding of how these architectures scale to handle production-grade, distributed data pipelines. I lack formal exposure to the massive parallelized optimization techniques, advanced loss-function tuning, and deployment paradigms required for industrial-scale generative models and multi-modal systems.

The Amazon ML Summer School represents the precise bridge I need to transition from an applied local developer to a scalable systems researcher. I want to anchor my empirical intuition in the mathematical rigor taught by Amazon’s scientists, particularly in themes like Large Language Models and advanced Deep Learning. My deep-rooted curiosity is paired with an established execution capability—I do not just read papers; I build, debug, and open-source their implementations. Immersing myself in this rigorous curriculum alongside a high-caliber cohort will provide the critical constraints and insights necessary to refine my research methodology and prepare me for impactful contributions to the field of machine learning.

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